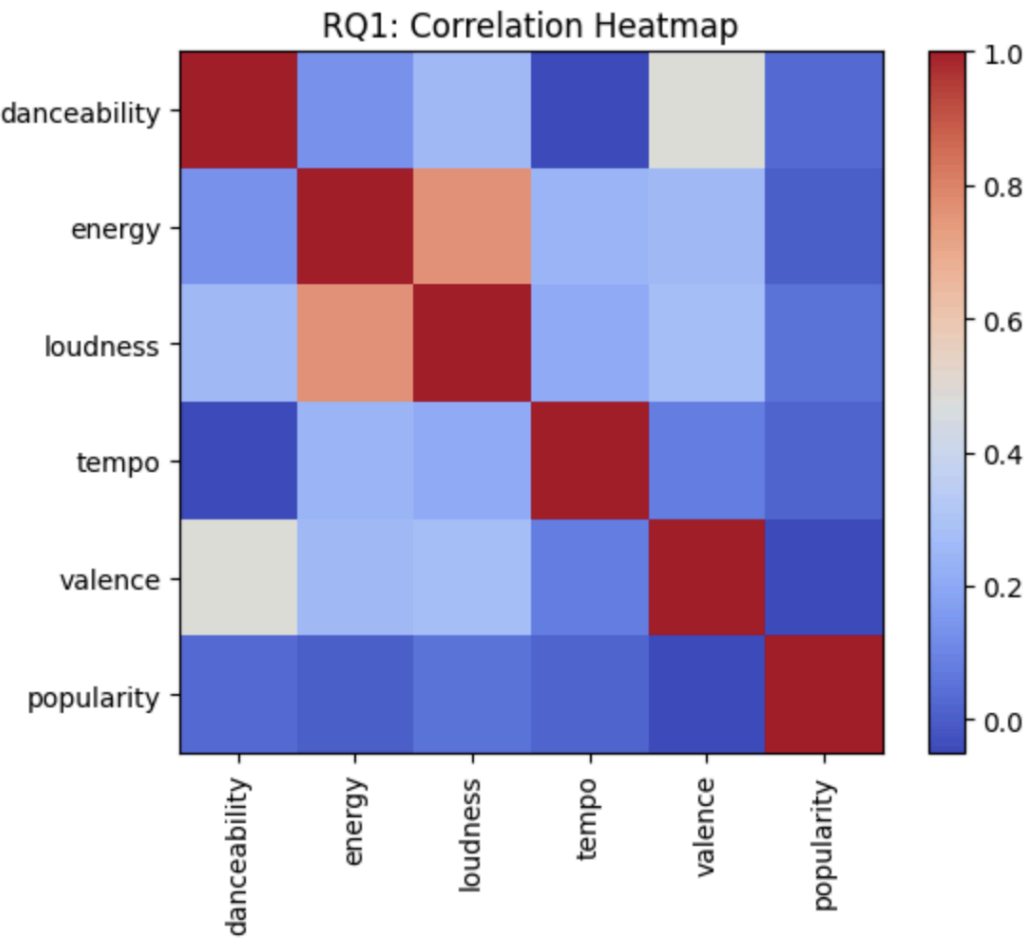


SPOTIFY SONG POPULARITY PREDICTION USING MACHINE LEARNING

RQ1: Correlation Table

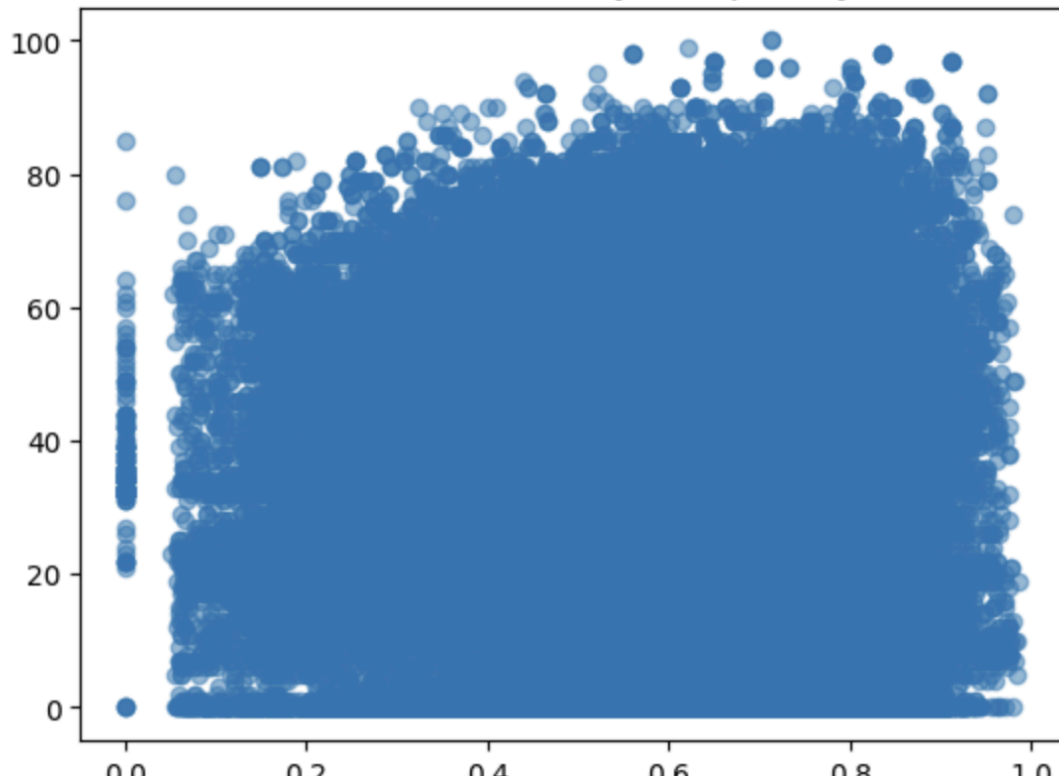
	danceability	energy	loudness	tempo	valence	popularity
danceability	1.000000	0.134325	0.259076	-0.050448	0.477347	0.035444
energy	0.134325	1.000000	0.761690	0.247852	0.258937	0.001053
loudness	0.259076	0.761690	1.000000	0.212447	0.279851	0.050420
tempo	-0.050448	0.247852	0.212447	1.000000	0.078269	0.013212
valence	0.477347	0.258937	0.279851	0.078269	1.000000	-0.040522
popularity	0.035444	0.001053	0.050420	0.013212	-0.040522	1.000000



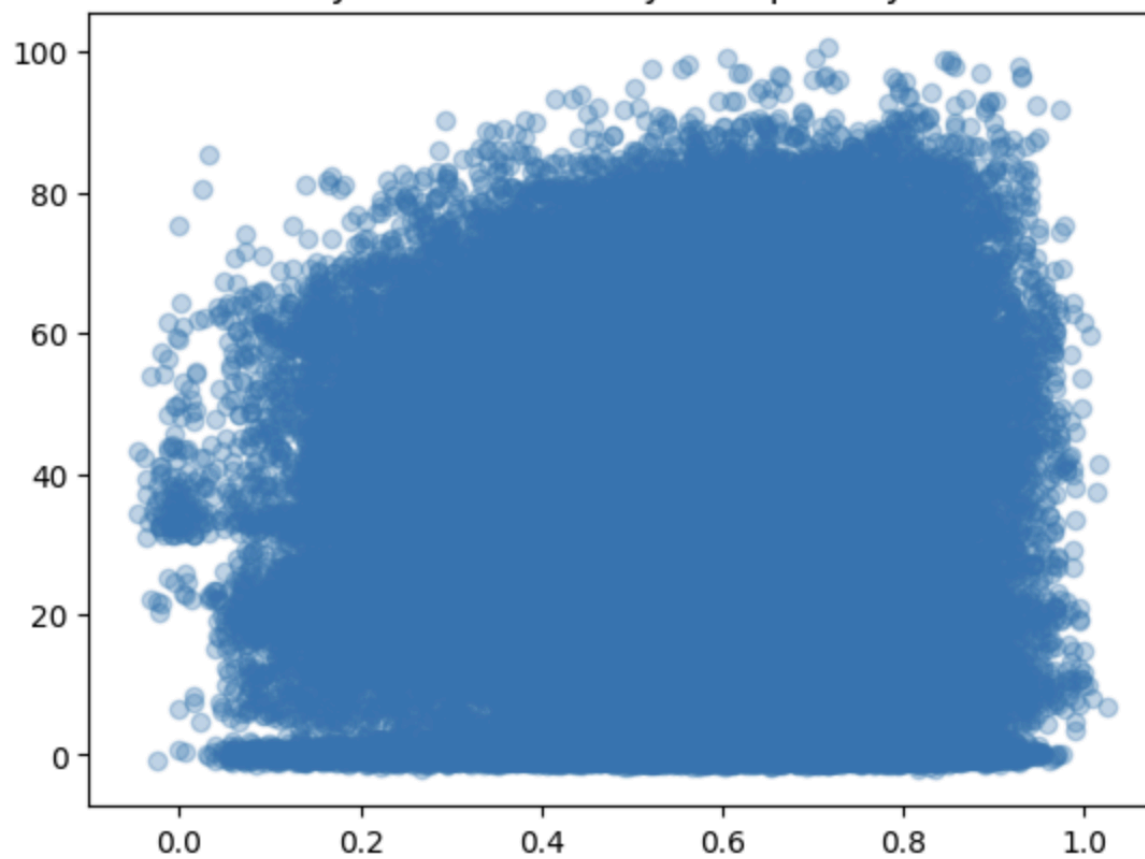
RQ2: Danceability vs Popularity

	danceability	popularity
count	113999.000000	113999.000000
mean	0.566801	33.238827
std	0.173543	22.304959
min	0.000000	0.000000
25%	0.456000	17.000000
50%	0.580000	35.000000
75%	0.695000	50.000000
max	0.985000	100.000000

Scatter: Danceability vs Popularity

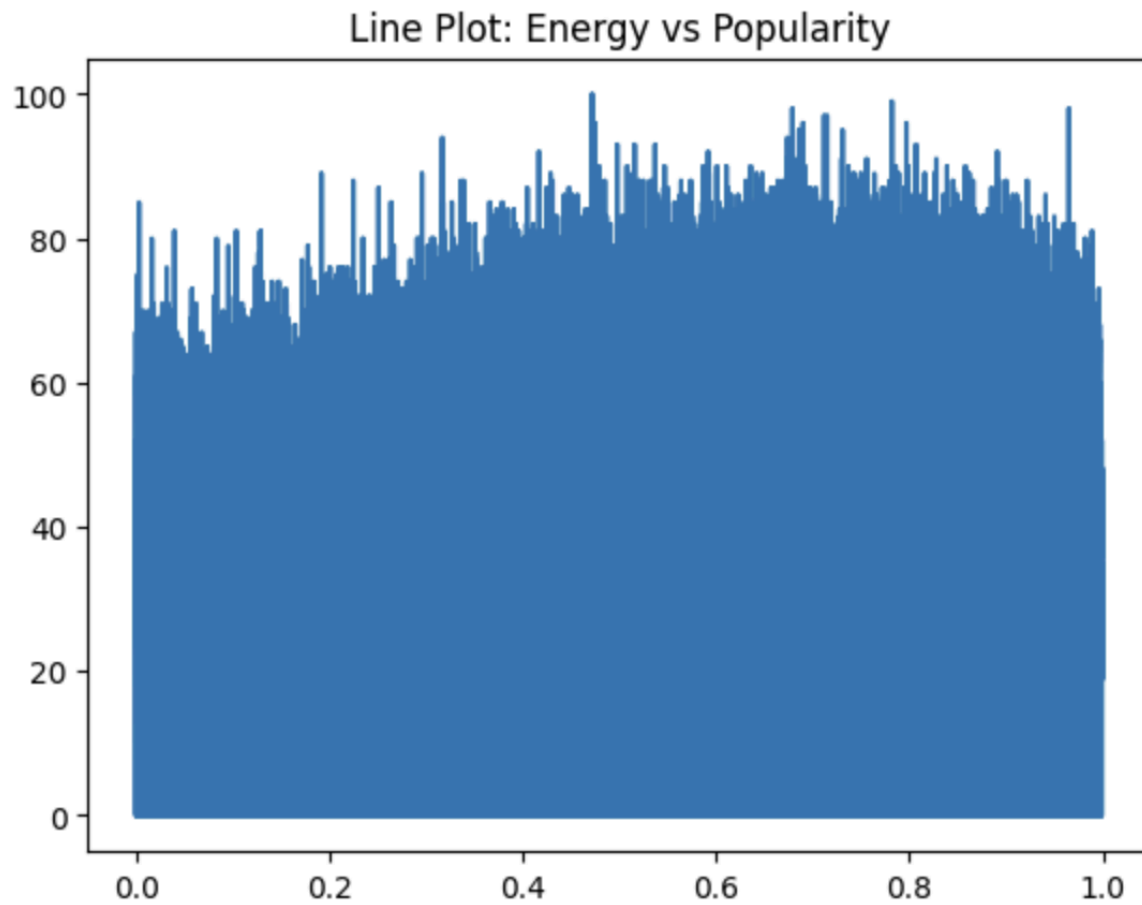


jitter: Danceability vs Popularity

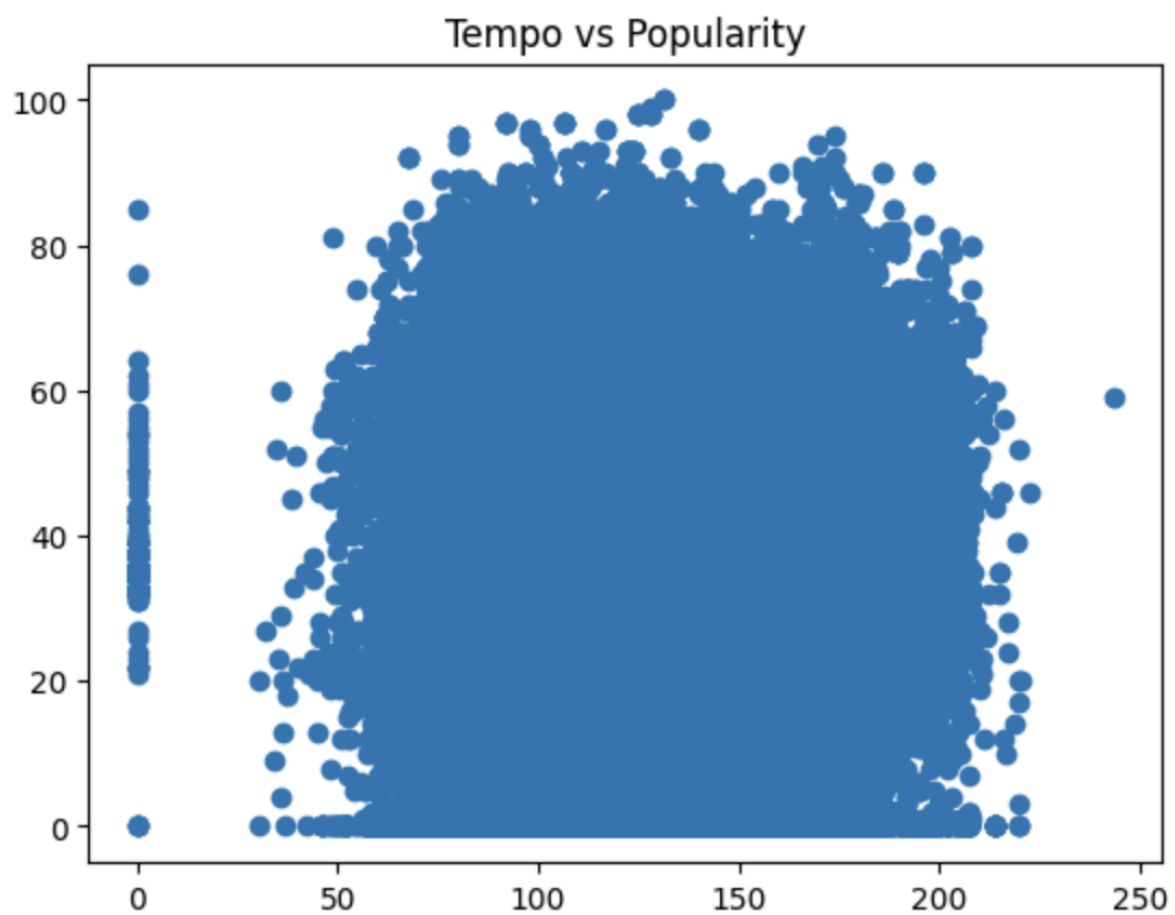


RQ3: Energy vs Popularity

	energy	popularity
count	113999.000000	113999.000000
mean	0.641383	33.238827
std	0.251530	22.304959
min	0.000000	0.000000
25%	0.472000	17.000000
50%	0.685000	35.000000
75%	0.854000	50.000000
max	1.000000	100.000000

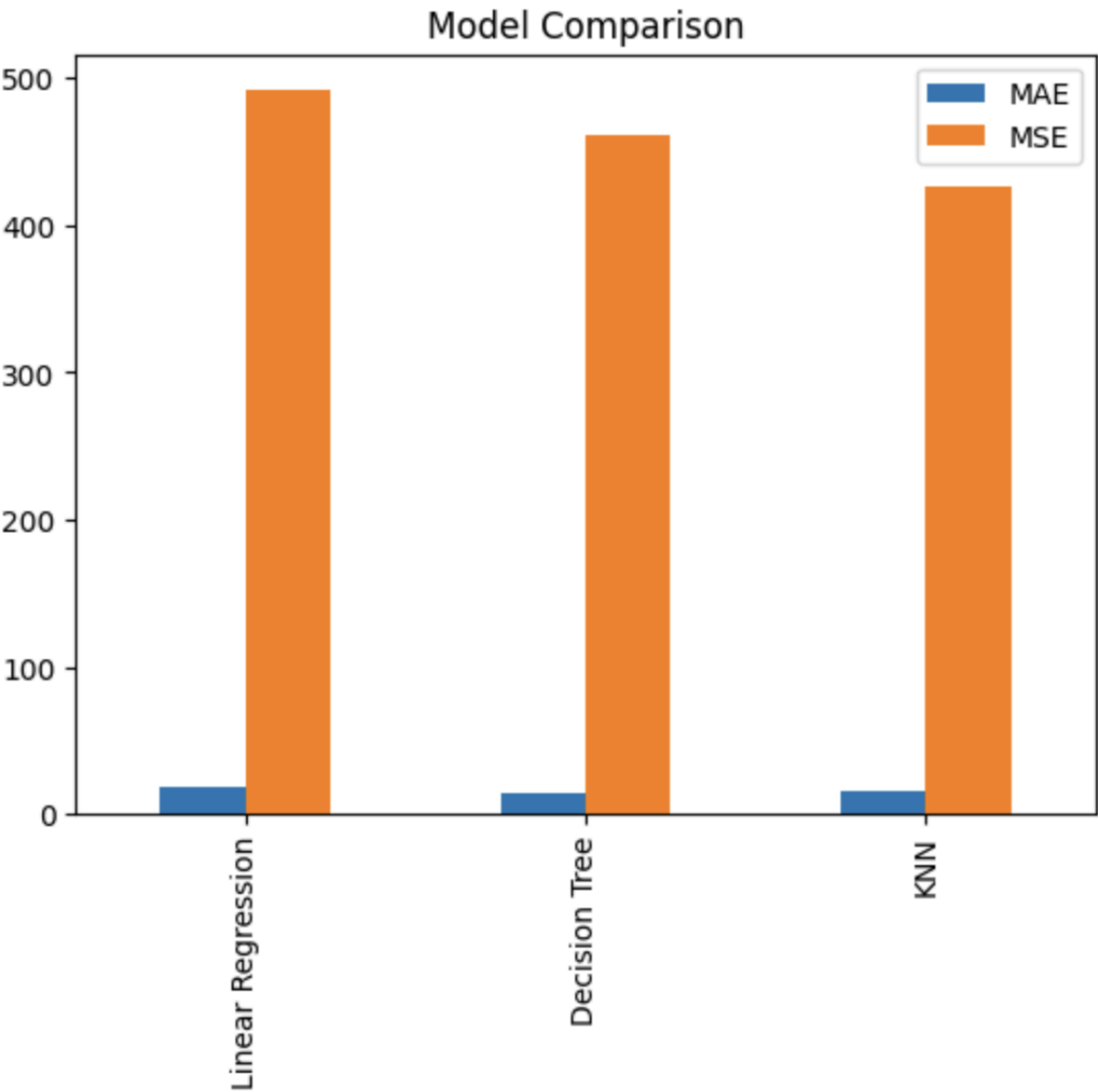


RQ4: Tempo vs Popularity



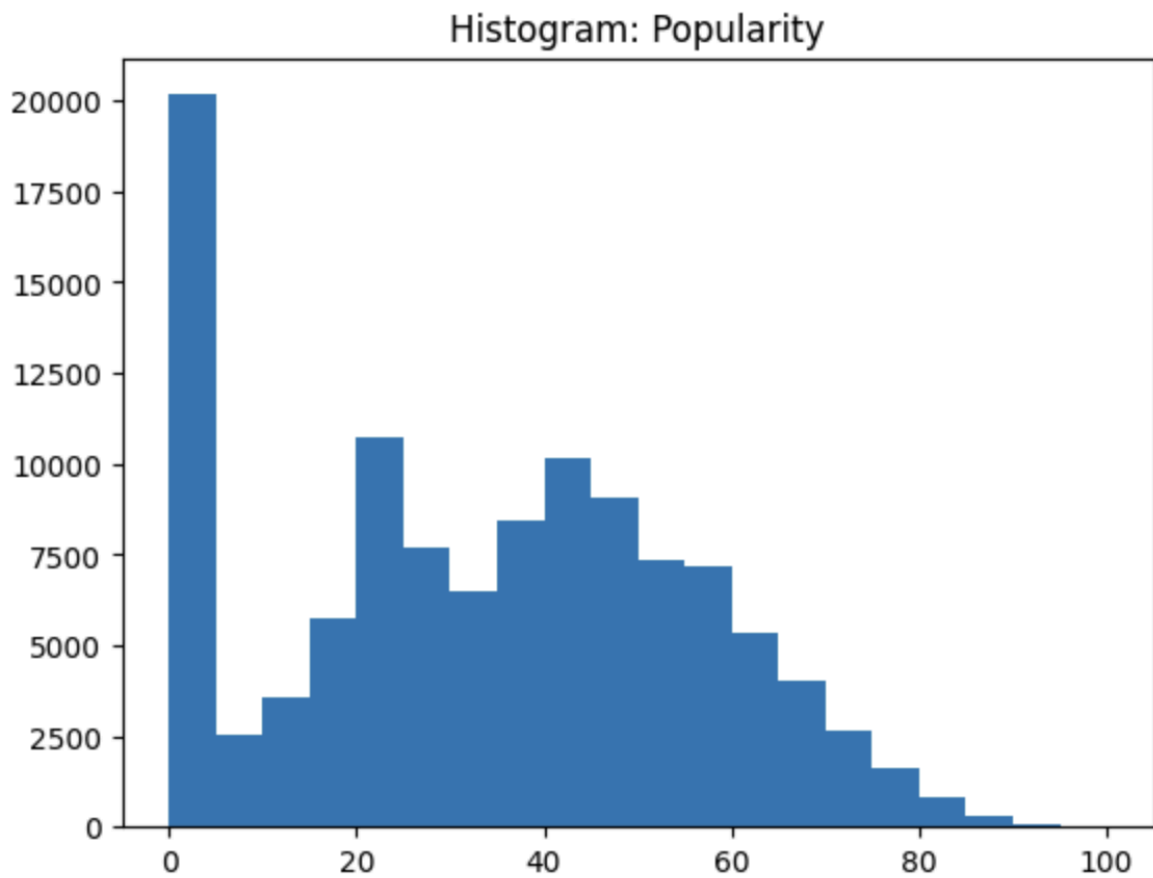
RQ5: Model Comparison Table

	Model	MAE	MSE
0	Linear Regression	18.666725	491.322358
1	Decision Tree	13.776615	461.322997
2	KNN	15.870202	425.631423



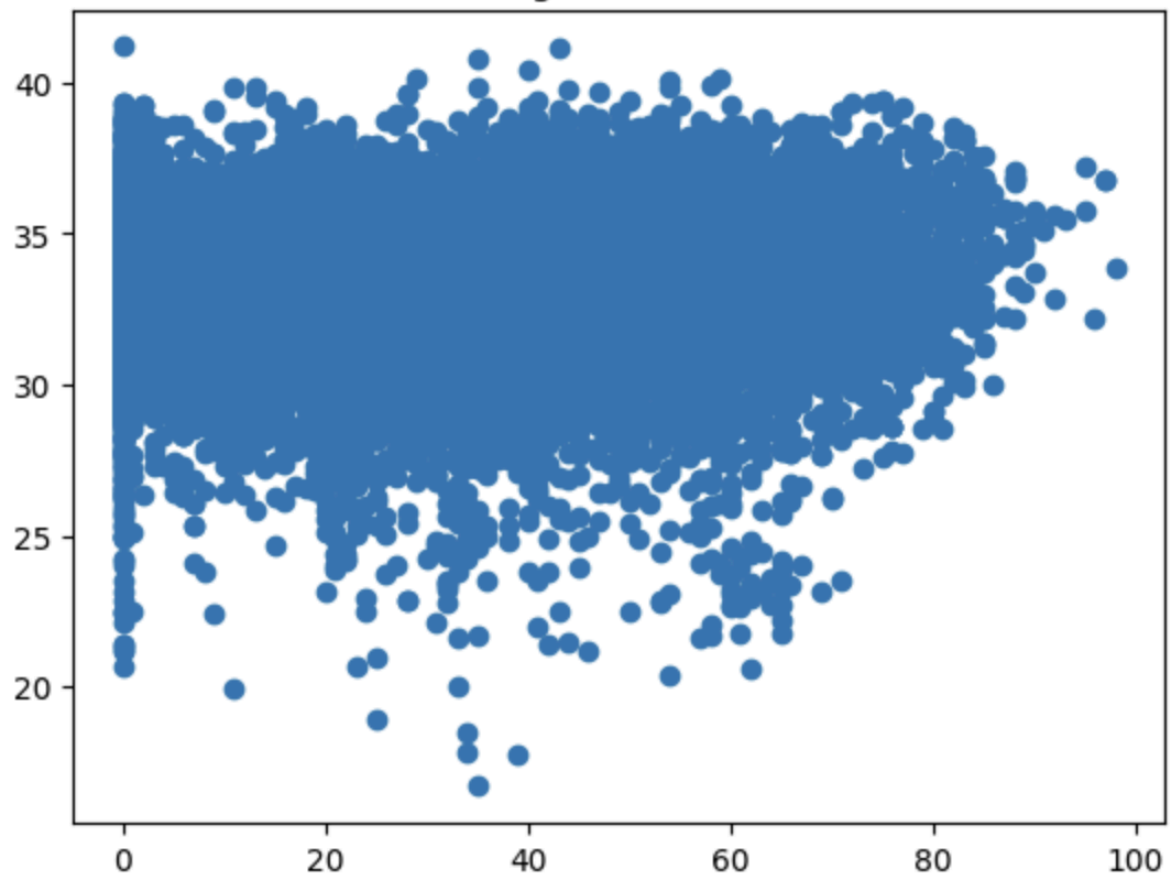
RQ6: Popularity Distribution

```
count    113999.000000
mean      33.238827
std       22.304959
min        0.000000
25%       17.000000
50%       35.000000
75%       50.000000
max       100.000000
Name: popularity, dtype: float64
```



RQ7: Prediction Results

Linear Regression Prediction



Decision Tree Prediction

